



2013 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 1

Tuesday 26th February 2013

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 116 boys

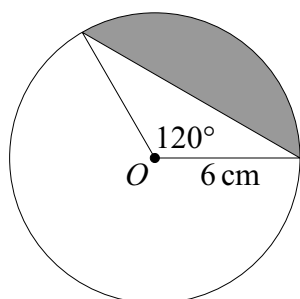
Examiner
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SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

1



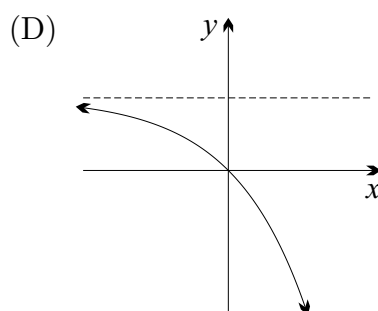
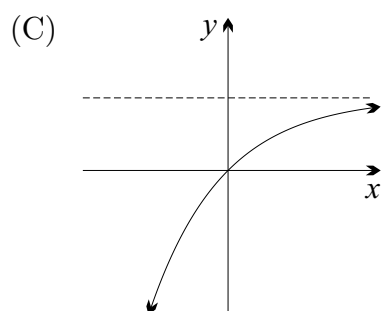
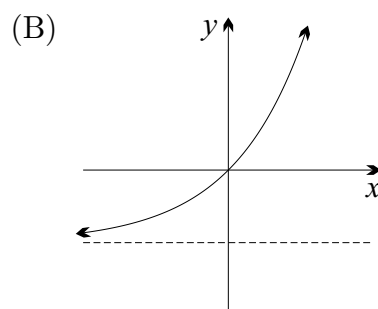
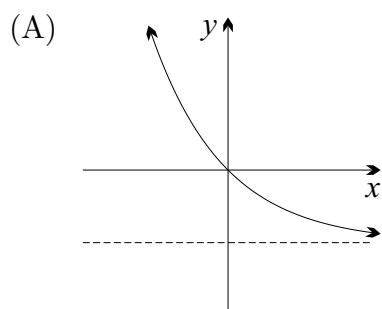
The diagram above shows a shaded segment which subtends an angle of 120° at the centre of a circle with radius 6 cm. The area of the segment correct to one decimal place is:

- (A) 22.1 cm^2 (B) 37.7 cm^2 (C) 2144.4 cm^2 (D) 2160.0 cm^2

QUESTION TWO

1

Which of the following graphs best represents $y = 1 - e^x$?



QUESTION THREE

The derivative of $\log(1 + x^2)$ is:

1

- (A) $\frac{2}{1+x}$ (B) $\frac{1}{1+x^2}$ (C) $\frac{2x}{1+x^2}$ (D) $\frac{x}{1+x^2}$

QUESTION FOUR

The expression $\cos^2 x$ is equivalent to:

1

- (A) $1 + \cos 2x$ (B) $1 - \cos 2x$
(C) $\frac{1}{2}(1 + \cos 2x)$ (D) $\frac{1}{2}(1 - \cos 2x)$

QUESTION FIVE

The focal length of the parabola with equation $y = 2x^2$ is:

1

- (A) 8 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{8}$

QUESTION SIX

The volume V cubic centimetres of a sphere with radius r centimetres is $V = \frac{4}{3}\pi r^3$.
The radius is increasing at a rate of 3 cm/s. At what rate is V increasing?

1

- (A) $\frac{4}{3}\pi r^3$ (B) $4\pi r^3$ (C) $4\pi r^2$ (D) $12\pi r^2$

QUESTION SEVEN

The derivative of $\tan^{-1} 2x$ is:

1

- (A) $\frac{2}{4+x^2}$ (B) $\frac{2}{1+4x^2}$ (C) $\frac{1}{4+x^2}$ (D) $\frac{1}{1+4x^2}$

QUESTION EIGHT

The expression $\cos x + \sin x$ is equivalent to:

1

- (A) $\sqrt{2}\cos(x + \frac{\pi}{4})$ (B) $\sqrt{2}\cos(x - \frac{\pi}{4})$
(C) $\sqrt{2}\cos(x + \frac{3\pi}{4})$ (D) $\sqrt{2}\cos(x - \frac{3\pi}{4})$

QUESTION NINE

The directrix of the parabola with equation $(y - 1)^2 = 12(x + 1)$ is given by:

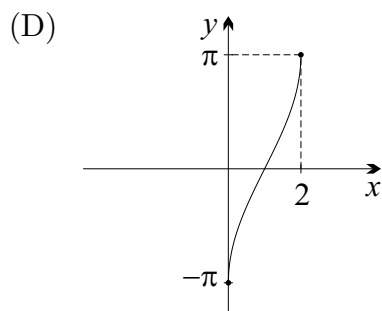
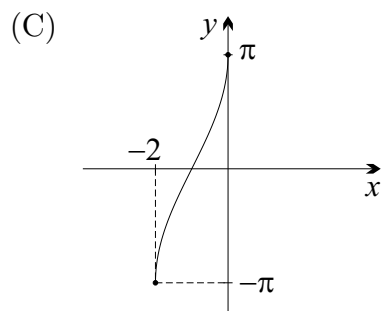
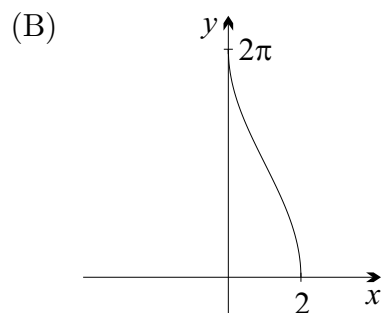
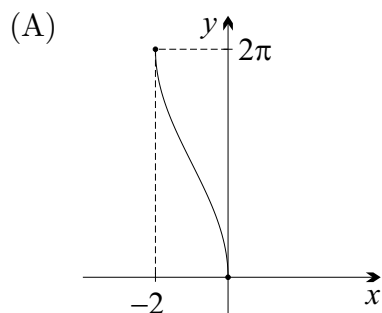
1

- (A) $x = -4$ (B) $x = -2$ (C) $x = 2$ (D) $x = 4$

QUESTION TEN

Which of the following graphs best represents $y = 2 \sin^{-1}(x + 1)$?

1



————— End of Section I —————

Exam continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet. **Marks**

(a) Simplify:

(i) $\ln(e^3)$

1

(ii) $\tan\left(\frac{5\pi}{6}\right)$

1

(iii) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

1

(b) Differentiate:

(i) $\tan 3x$

1

(ii) $x \sin^{-1} x$

2

(c) Evaluate $\lim_{x \rightarrow 0} \frac{\tan x}{2x}$.

1

(d) Prove that $\frac{\sin 2A}{\cos 2A - 1} = -\cot A$.

3

(e) An equilateral triangle has sides of length ℓ .

(i) Show that the area A of the triangle is given by $A = \frac{\sqrt{3}}{4}\ell^2$.

1

(ii) The area of the triangle is increasing at the rate of $9 \text{ cm}^2/\text{min}$. Determine the rate at which ℓ is increasing when the sides are 6 cm.

2

(f) Write down the general solution of $\sin x = \frac{1}{2}$.

2

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) (i) Differentiate $y = \cos 2x$. **1**

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \sin x \cos x \, dx$. **2**

(b) Evaluate $\int_0^4 \frac{x}{\sqrt{9+x^2}} \, dx$. **2**

(c) Use the substitution $t = \tan \frac{1}{2}\theta$ to show that **2**

$$\frac{1 + \sin \theta}{1 + \cos \theta} = \frac{1}{2}(1 + \tan \frac{1}{2}\theta)^2.$$

(d) The acute angle between the line $4x + 3y = 8$ and the line $ax + by = 8$ is 45° . **3**

Find the possible values of the fraction $\frac{a}{b}$.

(e) (i) Sketch $y = \cos x + 1$ for $-\pi \leq x \leq \pi$. **1**

(ii) The region bounded by $y = \cos x + 1$ and the x -axis, where $-\pi \leq x \leq \pi$, is rotated about the x -axis to generate a solid. Find the volume of this solid. **4**

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

(a) Determine $\int \frac{1}{\sqrt{3-4x^2}} \, dx$. **2**

(b) The point $P(6t, 3t^2)$ lies on the parabola $x^2 = 12y$.

(i) Show that the equation of the normal at P is $x + ty = 6t + 3t^3$. **2**

(ii) The normal at P cuts the y -axis at A .

(α) The mid-point of PA is R . Find the coordinates of R in terms of t . **1**

(β) The locus of R is another parabola. Find its vertex and focal length. **2**

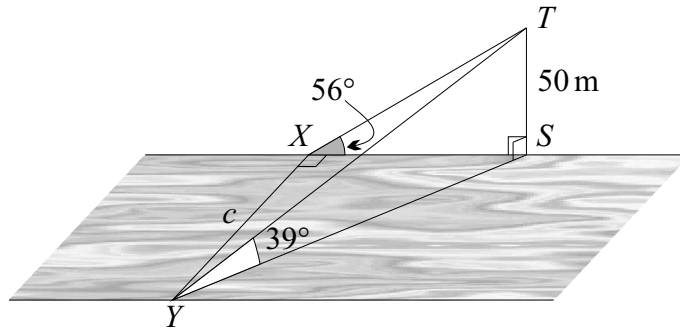
(c) Solve $2 \cos^2 x + \sqrt{3} \sin 2x = 0$ for $0 \leq x \leq 2\pi$. **3**

(d) (i) Express $\sqrt{3} \sin \theta - \cos \theta$ in the form $R \sin(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. **2**

(ii) Hence solve $\sqrt{3} \sin \theta - \cos \theta \geq 1$ for $0 \leq \theta \leq 2\pi$. **3**

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)

3

From a point X on a straight canal bank a surveyor measures the angle of elevation to T , the top of a 50 m tower with base S on the same bank. This angle is 56° . Directly opposite X on the other side of the canal, another surveyor at Y finds the angle of elevation to T is 39° . The points X , Y and S are on level ground.

Find c , the width of the canal, correct to three significant figures.

(b) (i) Prove by mathematical induction that for all integers $n \geq 1$ **3**

$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}.$$

(ii) Hence evaluate

1

$$\lim_{n \rightarrow \infty} \left(\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} \right).$$

(c) The function $\sin^{-1} x$ is odd. By finding x as a function of y or otherwise, show algebraically that**2**

$$y = 4 \cos^{-1} x - 2\pi$$

is also odd.

(d) Suppose that

$$\int_{-2h}^{2h} f(x) dx = A \times f(-h) + B \times f(0) + C \times f(h) \quad (**)$$

for $f(x) = 1$, $f(x) = x$ and $f(x) = x^2$.

(i) Determine the values of A , B and C in terms of h .**4**(ii) Show that equation $(**)$ is also valid when $f(x) = x^3$.**1**(iii) Equation $(**)$ may be used to approximate the integrals of other functions. Use it to show that**1**

$$\int_{-1}^1 2^x dx \doteq \frac{2}{3}(3\sqrt{2} - 1).$$

End of Section II

END OF EXAMINATION

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

FORM VI MATHEMATICS EXTENSION 1

Q1/ A ✓

Q2/ D ✓

Q3/ C ✓

Q4/ C ✓

Q5/ D ✓

Q6/ D ✓

Q7/ B ✓

Q8/ B ✓

Q9/ A ✓

Q10/ C ✓

Q11/

a) i) 3 ✓

ii) $-\frac{1}{\sqrt{3}}$ ✓

iii) $\frac{5\pi}{6}$ ✓

b) i) $3\sec^2 3x$ ✓

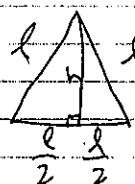
ii) $\sin^{-1} x + \frac{x}{\sqrt{1-x^2}}$ ✓

c) $\frac{1}{2}$ ✓

$$\begin{aligned} \text{d) LHS} &= \frac{2\sin A \cos A}{1 - 2\sin^2 A - 1} \quad \checkmark \\ &= \frac{2\sin A \cos A}{-2\sin^2 A} \quad \checkmark \\ &= -\frac{\cos A}{\sin A} \\ &= -\cot A \quad \checkmark \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} \text{e) i) } A &= \frac{1}{2} l^2 \sin \frac{\pi}{3} \\ &= \frac{\sqrt{3}}{4} l^2 \quad \checkmark \end{aligned}$$

OR



$$h^2 = l^2 - \left(\frac{l}{2}\right)^2$$

$$h^2 = \frac{3l^2}{4}$$

$$h = \frac{l\sqrt{3}}{2}$$

$$\therefore A = \frac{1}{2} l \cdot \frac{l\sqrt{3}}{2} = \frac{\sqrt{3}}{4} l^2$$

$$\text{ii)} \quad \frac{dA}{dt} = 9, \quad \frac{dA}{dl} = \frac{\sqrt{3}}{2} l, \quad \frac{dl}{dt} = ?$$

$$\frac{dA}{dt} = \frac{dA}{dl} \cdot \frac{dl}{dt}$$

$$\therefore 9 = \frac{\sqrt{3}}{2} l \cdot \frac{dl}{dt} \quad \checkmark$$

$$l = 6, \quad 9 = \frac{\sqrt{3}}{2} \cdot 6 \cdot \frac{dl}{dt}$$

$$\frac{dl}{dt} = \frac{9}{3\sqrt{3}}$$

$$= \frac{3}{\sqrt{3}}$$

$$= \sqrt{3} \text{ cm/min} \quad \checkmark$$

$$\text{f)} \quad \sin x = \frac{1}{2}$$

$$\therefore x = \frac{\pi}{6} + 2n\pi \quad \checkmark \quad \text{or} \quad \frac{5\pi}{6} + 2n\pi \quad \checkmark$$

for integers n .

OR

$$x = (-1)^n \cdot \frac{\pi}{6} + n\pi$$

for integers n .

Q12/

$$\text{a) i)} \quad y = \cos 2x$$

$$\frac{dy}{dx} = -2 \sin 2x \quad \checkmark$$

$$= -4 \sin x \cos x$$

$$\text{ii)} \quad \int_0^{\pi/2} \sin x \cos x \, dx = -\frac{1}{4} \int_0^{\pi/2} -4 \sin x \cos x \, dx$$

$$= -\frac{1}{4} [\cos 2x]_0^{\pi/2} \quad \checkmark$$

$$= -\frac{1}{4} (\cos \pi - \cos 0)$$

$$= -\frac{1}{4} (-1 - 1)$$

$$= \frac{1}{2} \quad \checkmark$$

$$\text{b)} \quad \frac{d}{dx} (9+x^2)^{1/2} = \frac{1}{2} (9+x^2)^{-1/2} \cdot 2x$$

$$= \frac{x}{\sqrt{9+x^2}}$$

$$\therefore \int_0^4 \frac{x}{\sqrt{9+x^2}} \, dx = \left[(9+x^2)^{1/2} \right]_0^4 \quad \checkmark$$

$$= \sqrt{25} - \sqrt{9} \quad \checkmark$$

$$= 2 \quad \checkmark$$

$$c) t = \tan \frac{\theta}{2}$$

$$\text{so, } \sin \theta = \frac{2t}{1+t^2}, \quad \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\text{LHS} = \frac{1 + \frac{2t}{1+t^2}}{1 + \frac{1-t^2}{1+t^2}}$$

$$= \frac{1+t^2+2t}{1+t^2+1-t^2}$$

$$= \frac{(1+t)^2}{2}$$

$$= \frac{1}{2} (1 + \tan \frac{\theta}{2})^2$$

$$= \text{RHS}$$

$$d) m_1 = -\frac{4}{3}, \quad m_2 = -\frac{a}{b}$$

$$\therefore \tan 45^\circ = \left| \frac{-\frac{4}{3} + \frac{a}{b}}{1 + \frac{4a}{3b}} \right|$$

$$1 = \left| \frac{-4b+3a}{3b+4a} \right|$$

$$\frac{-4b+3a}{3b+4a} = 1$$

$$-4b+3a = 3b+4a$$

$$-a = 7b$$

$$\frac{a}{b} = -7 \quad \checkmark \quad \text{OR}$$

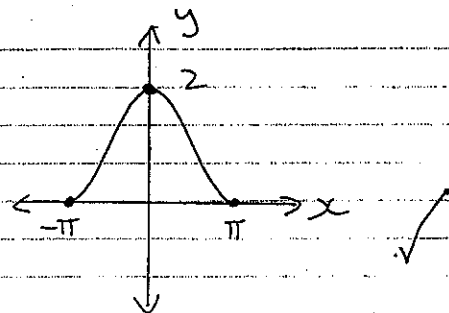
$$\frac{-4b+3a}{3b+4a} = -1$$

$$-4b+3a = -3b-4a$$

$$7a = b$$

$$\frac{a}{b} = \frac{1}{7} \quad \checkmark$$

e) i)



$$ii) V = \pi \int_{-\pi}^{\pi} (\cos x + 1)^2 dx \quad \checkmark$$

$$= 2\pi \int_{-\pi}^{\pi} (\cos^2 x + 2\cos x + 1) dx$$

$$= 2\pi \int_{-\pi}^{\pi} \frac{1}{2} (1 + \cos 2x) dx + 2\pi \int_{-\pi}^{\pi} (2\cos x + 1) dx \quad \checkmark$$

$$= \pi \left[x + \frac{1}{2} \sin 2x \right]_{-\pi}^{\pi} + 2\pi \left[\sin x + x \right]_{-\pi}^{\pi} \quad \checkmark$$

$$= \pi (\pi + 0) + 2\pi (0 + \pi)$$

$$= \pi^2 + 2\pi^2$$

$$= 3\pi^2 \text{ units}^3 \quad \checkmark$$

Q13

$$a) \int \frac{1}{\sqrt{3-4x^2}} dx = \frac{1}{2} \int \frac{1}{\sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 - x^2}} dx \checkmark$$

$$= \frac{1}{2} \sin^{-1}\left(\frac{2x}{\sqrt{3}}\right) + C \checkmark$$

$$b) i) y = \frac{x^2}{12}$$

$$\frac{dy}{dx} = \frac{x}{6}$$

$$x = 6t, \quad \frac{dy}{dx} = t$$

$\therefore$ gradient of normal is $-\frac{1}{t} \checkmark$

$$m = -\frac{1}{t} \quad P(6t, 3t^2)$$

$$\therefore y - 3t^2 = -\frac{1}{t}(x - 6t) \checkmark$$

$$ty - 3t^3 = -x + 6t \checkmark$$

$$x + ty = 6t + 3t^3$$

$$ii) a) A(0, 6 + 3t^2) \quad P(6t, 3t^2)$$

$$\therefore R(3t, 3 + 3t^2) \checkmark$$

$$b) x = 3t \quad y = 3 + 3t^2$$

$$3y = 9 + 9t^2$$

$$3y = 9 + x^2$$

$$x^2 = 3(y - 3) \checkmark$$

$$\therefore V(0, 3), \quad a = \frac{3}{4} \checkmark \text{ for both}$$

$$c) 2\cos^2 x + \sqrt{3}\sin 2x = 0 \text{ for } 0 \leq x \leq 2\pi$$

$$2\cos^2 x + 2\sqrt{3}\sin x \cos x = 0$$

$$2\cos x (\cos x + \sqrt{3}\sin x) = 0$$

$$\therefore \cos x = 0 \text{ or } \cos x + \sqrt{3}\sin x = 0$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2} \checkmark$$

$$\sqrt{3}\sin x = -\cos x$$

$$\tan x = -\frac{1}{\sqrt{3}}$$

Quad II, IV

$$[x = \frac{\pi}{6}]$$

$$x = \frac{5\pi}{6}, \frac{11\pi}{6} \checkmark$$

$$\therefore x = \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6} \text{ for } 0 \leq x \leq 2\pi$$

$$d) i) \sqrt{3}\sin \theta - \cos \theta \equiv R \sin(\theta - \alpha)$$

$$= R \sin \theta \cos \alpha - R \sin \alpha \cos \theta$$

By equating coeff;

$$R \cos \alpha = \sqrt{3} \quad \text{and} \quad R \sin \alpha = 1 \checkmark$$

$$\therefore R = 2, \quad \alpha = \frac{\pi}{6} \checkmark$$

$$\therefore \sqrt{3}\sin \theta - \cos \theta \equiv 2 \sin(\theta - \frac{\pi}{6})$$

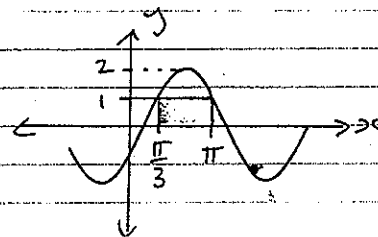
$$ii) \sqrt{3}\sin \theta - \cos \theta \geq 1 \checkmark$$

$$2 \sin(\theta - \frac{\pi}{6}) \geq 1$$

$$\sin(\theta - \frac{\pi}{6}) \geq \frac{1}{2}$$

$$\text{for equality, } \theta - \frac{\pi}{6} = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

$$\therefore \theta = \frac{\pi}{3} \text{ or } \pi \checkmark$$



$$\therefore \frac{\pi}{3} \leq \theta \leq \pi \checkmark$$

Q14/

a) $XS = \frac{50}{\tan 56^\circ}$ $YS = \frac{50}{\tan 39^\circ}$ ✓

$$\therefore c^2 = \left(\frac{50}{\tan 39^\circ}\right)^2 - \left(\frac{50}{\tan 56^\circ}\right)^2$$

$$= 2675.023 \dots$$

$$\therefore c = 51.7 \text{ (1.d.p.)}$$
 ✓

b) i) ① when $n=1$

$$LHS = \frac{1}{1 \times 5}$$

$$= \frac{1}{5}$$

$$RHS = \frac{1}{4 \times 1 + 1}$$

$$= \frac{1}{5}$$

$$= LHS$$

$\therefore$ true when $n=1$ ✓

② Assume true for $n=k$

$$\text{i.e. } \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$$

③ Prove true for $n=k+1$

$$\text{R.T.P. } \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4k+1)(4k+5)}$$

$$= \frac{k+1}{4k+5}$$

$$LHS = \left(\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \dots + \frac{1}{(4k-3)(4k+1)} \right) + \frac{1}{(4k+1)(4k+5)}$$

$$= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)}$$
 ✓

$$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$$

$$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)}$$

$$= \frac{k+1}{4k+5}$$

$$= RHS$$
 ✓

As true for $n=k+1$ and true for $n=1$
it follows that it is true for $n=2, 3, 4, \dots$
and all integers of n greater or
equal to 1

ii) $\lim_{n \rightarrow \infty} \frac{n}{4n+1} = \lim_{n \rightarrow \infty} \frac{1}{4 + \frac{1}{n}}$ ✓

$$= \frac{1}{4}$$

c)

$$y = 4 \cos^{-1} x - 2\pi$$

$$4 \cos^{-1} x = y + 2\pi$$

$$\cos^{-1} x = \frac{y}{4} + \frac{\pi}{2}$$

$$x = \cos \left(\frac{y}{4} + \frac{\pi}{2} \right)$$
 ✓

$$= -\sin \left(\frac{y}{4} \right)$$

$$\text{So } \sin \left(\frac{y}{4} \right) = -x$$

$$y = 4 \sin^{-1}(-x)$$

$$= -4 \sin^{-1} x$$
 ✓

hence y is odd as $\sin^{-1} x$ is odd

d) i) * $f(x) = 1$ gives

$$\int_{-2h}^{2h} 1 dx = A + B + C$$

$$\therefore A + B + C = 4h \quad \checkmark$$

* $f(x) = x$ gives

$$\int_{-2h}^{2h} x dx = -Ah + Ch$$

$$-Ah + Ch = \left[\frac{x^2}{2} \right]_{-2h}^{2h}$$

$$\therefore A = C = 0 \quad \checkmark$$

* $f(x) = x^2$ gives

$$\int_{-2h}^{2h} x^2 dx = Ah^2 + Ch^2$$

$$\left[\frac{x^3}{3} \right]_{-2h}^{2h} = Ah^2 + Ch^2$$

$$\frac{16h^3}{3} = Ah^2 + Ch^2 \quad \checkmark$$

Using $A = C$, $2Ah^2 = \frac{16h^3}{3}$

$$A = \frac{8h}{3}$$

$$C = \frac{8h}{3}$$

and ~~and~~ $B + \frac{16h}{3} = 4h$

$$B = -\frac{4h}{3}$$

$$\therefore A = \frac{8h}{3}, B = -\frac{4h}{3}, C = \frac{8h}{3} \quad \checkmark$$

ii) $f(x) = x^3$

$$\text{LHS} = \int_{-2h}^{2h} x^3 dx$$

$$= \left[\frac{x^4}{4} \right]_{-2h}^{2h}$$

$$\begin{aligned} \text{RHS} &= 0 \\ &= -Ah^3 + Ch^3 \\ &= h^3(-A+C) \\ &= 0 \\ &= \text{LHS} \quad \checkmark \end{aligned}$$

iii) $h = \frac{1}{2}$, $f(x) = 2^x$

$$\therefore \int_{-1}^1 2^x dx = A \cdot 2^{-\frac{1}{2}} + B \cdot 2^0 + C \cdot 2^{\frac{1}{2}}$$

$$= \frac{4}{3} \frac{1}{\sqrt{2}} - \frac{2}{3} + \frac{4}{3} \sqrt{2}$$

$$= \frac{2}{3} \left(\frac{2}{\sqrt{2}} - 1 + 2\sqrt{2} \right) \quad \checkmark$$

$$= \frac{2}{3} (3\sqrt{2} - 1)$$